

ChiSquare_Binary_1

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2025-03-04

Perspective Understanding

```
library(tidyverse)

## -- Attaching packages ----- tidyverse 1.3.2 --
## v ggplot2 3.4.0      v purrr  1.0.2
## v tibble  3.2.1      v dplyr  1.1.4
## v tidyr   1.3.0      v stringr 1.5.1
## v readr   2.1.5      v forcats 0.5.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()

df <- read.csv("/home/zzeng1/Innovation80/Innovation80/Perspective - Sheet1.csv")

# Convert empty strings to NA
df[df == ""] <- NA

# Remove rows with missing values in "Perspective Understanding"
df_cleaned <- df %>% drop_na(Perspective.Understanding)

df_cleaned <- df_cleaned |> filter(Participant.Type == "Senior")

# Standardize the text format for categorical variables
df_cleaned$Perspective.Understanding <- tolower(trimws(df_cleaned$Perspective.Understanding))

# Create a contingency table
contingency_table <- table(df_cleaned$Survey.Type, df_cleaned$Perspective.Understanding)

# Perform the Chi-Square Test
chi_test <- chisq.test(contingency_table)

## Warning in chisq.test(contingency_table): Chi-squared approximation may be
## incorrect

# Print results
print(chi_test)

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 1.3652, df = 1, p-value = 0.2426
```

```

# Display contingency table
print(contingency_table)

##
##      no yes
## Post 34   2
## Pre  29   6

# Relevel Survey.Type so "Pre" appears before "Post"
df_cleaned$Survey.Type <- factor(df_cleaned$Survey.Type, levels = c("Pre", "Post"))

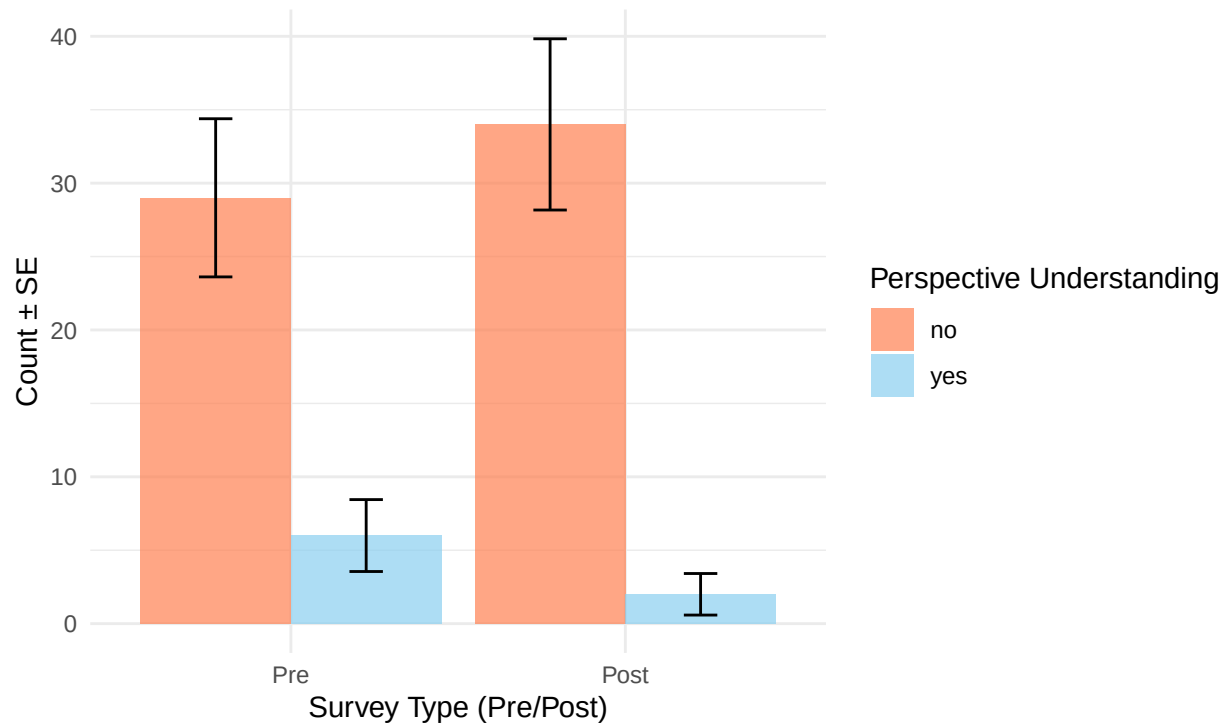
# Create a summary table with proportions and standard error (SE)
summary_df <- df_cleaned %>%
  group_by(Survey.Type, Perspective.Understanding) %>%
  summarise(Count = n(), .groups = "drop") %>%
  mutate(SE = sqrt(Count)) # Approximate standard error for count data

# Create the bar plot with vertical error bars and explanation note
ggplot(summary_df, aes(x = Survey.Type, y = Count, fill = Perspective.Understanding)) +
  geom_bar(stat = "identity", position = "dodge", alpha = 0.7) +
  geom_errorbar(aes(ymin = Count - SE, ymax = Count + SE),
               position = position_dodge(width = 0.9), width = 0.2) + # Adds vertical error bars
  scale_fill_manual(values = c("yes" = "#89CFF0", "no" = "#FF7F50")) +
  labs(title = "Perspective Understanding by Survey Type",
       subtitle = "'Yes' means the participant agrees that younger individuals
may not be able to understand their perspective; 'No' means they disagree.",
       x = "Survey Type (Pre/Post)",
       y = "Count ± SE",
       fill = "Perspective Understanding") +
  theme_minimal()

```

Perspective Understanding by Survey Type

'Yes' means the participant agrees that younger individuals may not be able to understand their perspective; 'No' means they disagree.



Disinterest in Interaction

```
# Read the CSV file
interaction_data <- read.csv("/home/zzeng1/Innovation80/Innovation80/Interaction - Sheet1.csv", stringsAsFactors = FALSE)
# Convert empty strings to NA and remove them
interaction_data[interaction_data == ""] <- NA

interaction_data <- interaction_data %>% drop_na(Disinterest.in.Interaction)

# Exclude data from "Strands of Wisdom"
interaction_data <- interaction_data %>% filter(Participant.Type == "Senior")
# Create a contingency table
contingency_table <- table(interaction_data$Survey.Type, interaction_data$Disinterest.in.Interaction)

# Perform Chi-Square Test
chi_test <- chisq.test(contingency_table)

# Print results
print(chi_test)

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 6.8674, df = 1, p-value = 0.008778
```

```
# Display contingency table
print(contingency_table)
```

```
##
##           No Yes
##   Post 35    1
##   Pre  26   10
```

A Pearson's chi-square test with Yates' continuity correction was conducted to examine the relationship between survey time point (Pre vs. Post) and the perception that the younger person was uninterested in getting to know someone older. The results indicated a statistically significant association, $X^2(1, N = [\text{sample size}]) = 6.87$, $p = .009$.

The significant result suggests that perceptions of younger individuals' disinterest in engaging with older individuals changed between the Pre and Post survey conditions. Specifically, participants were significantly less likely to perceive younger individuals as uninterested in getting to know older people in the Post survey compared to the Pre survey. This finding may indicate a positive shift in intergenerational engagement following the intervention. However, further research is needed to determine the extent and causes of this shift.

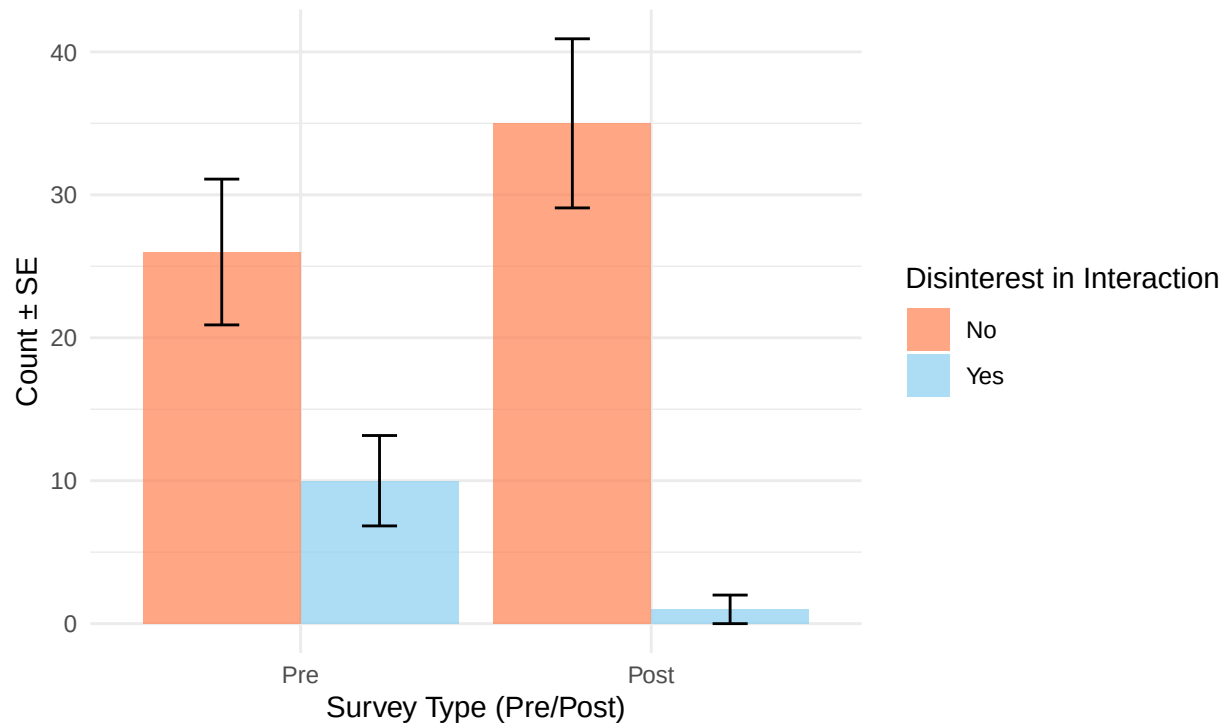
```
# Relevel Survey.Type so "Pre" appears before "Post"
interaction_data$Survey.Type <- factor(interaction_data$Survey.Type, levels = c("Pre", "Post"))

# Create a summary table with proportions and standard error (SE)
summary_data <- interaction_data %>%
  group_by(Survey.Type, Disinterest.in.Interaction) %>%
  summarise(Count = n(), .groups = "drop") %>%
  mutate(SE = sqrt(Count)) # Approximate standard error for count data

# Create the bar plot with custom colors and explanation note
ggplot(summary_data, aes(x = Survey.Type, y = Count, fill = Disinterest.in.Interaction)) +
  geom_bar(stat = "identity", position = "dodge", alpha = 0.7) +
  geom_errorbar(aes(ymin = Count - SE, ymax = Count + SE),
               position = position_dodge(width = 0.9), width = 0.2) +
  scale_fill_manual(values = c("Yes" = "#89CFF0", "No" = "#FF7F50")) +
  labs(title = "Disinterest in Interaction by Survey Type",
       subtitle = "'Yes' means the participant agrees that younger individuals
       might be uninterested in interactions; 'No' means they disagree.",
       x = "Survey Type (Pre/Post)",
       y = "Count ± SE",
       fill = "Disinterest in Interaction") +
  theme_minimal()
```

Disinterest in Interaction by Survey Type

'Yes' means the participant agrees that younger individuals might be uninterested in interactions; 'No' means they disagree.



```
# Read the dataset
inflexibility <- read.csv("/home/zzeng1/Innovation80/Innovation80/Inflexibility Perception - Sheet1.csv")

# Convert empty strings to NA and remove them
inflexibility[inflexibility == ""] <- NA
inflexibility <- inflexibility %>% drop_na(Inflexibility.Perception) # Adjust column name as needed

# Debugging: Check column names
print(colnames(inflexibility))

## [1] "Participant.Type"      "Program.Name"
## [3] "Survey.Type"          "Inflexibility.Perception"

# Ensure "Young" is correctly matched (trim spaces)
inflexibility$Participant.Type <- trimws(inflexibility$Participant.Type)

# Filter only Young participants
inflexibility <- inflexibility %>% filter(Participant.Type == "Young")

# Debug: Check if filtering removed all rows
print(nrow(inflexibility))

## [1] 73

print(head(inflexibility))

##   Participant.Type Program.Name Survey.Type Inflexibility.Perception
## 1             Young   West BLUES          Pre                    Yes
```

```

## 2          Young  West BLUES          Pre          No
## 3          Young  West BLUES          Pre          No
## 4          Young  West BLUES          Pre          No
## 5          Young  West BLUES          Pre          No
## 6          Young  West BLUES          Pre          Yes

# Create a contingency table
contingency_table <- table(inflexibility$Survey.Type, inflexibility$Inflexibility.Perception)

# Debug: Check if the table is empty
print(contingency_table)

##
##          No Yes
## Post 35    7
## Pre  19   12

if (nrow(contingency_table) == 0 | ncol(contingency_table) == 0) {
  print("Error: Contingency table is empty. Check column names and filtering.")
} else {
  # Perform Chi-Square Test
  chi_test <- chisq.test(contingency_table)

  # Print results
  print(chi_test)
}

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 3.4291, df = 1, p-value = 0.06406

# Relevel Survey.Type so "Pre" appears before "Post"
inflexibility$Survey.Type <- factor(inflexibility$Survey.Type, levels = c("Pre", "Post"))

# Calculate counts for each category
summary_data <- inflexibility %>%
  group_by(Survey.Type, Inflexibility.Perception) %>%
  summarise(Count = n(), .groups = "drop")

# Calculate standard error
summary_data <- summary_data %>%
  group_by(Survey.Type) %>%
  mutate(Standard_Error = sqrt(Count)) # Approximation assuming Poisson distribution

# Plot the bar chart with error bars and explanation note
ggplot(summary_data, aes(x = Survey.Type, y = Count, fill = Inflexibility.Perception)) +
  geom_bar(stat = "identity", position = position_dodge(), width = 0.6, alpha = 0.7) +
  geom_errorbar(aes(ymin = Count - Standard_Error, ymax = Count + Standard_Error),
    width = 0.2, position = position_dodge(0.6)) +
  scale_fill_manual(values = c("Yes" = "#89CFF0", "No" = "#FF7F50")) +
  labs(title = "Inflexibility Perception (Pre vs Post)",
    subtitle = "'Yes' means agreeing that older individuals
    may be inflexible; 'No' means disagreement.",
    x = "Survey Type",

```

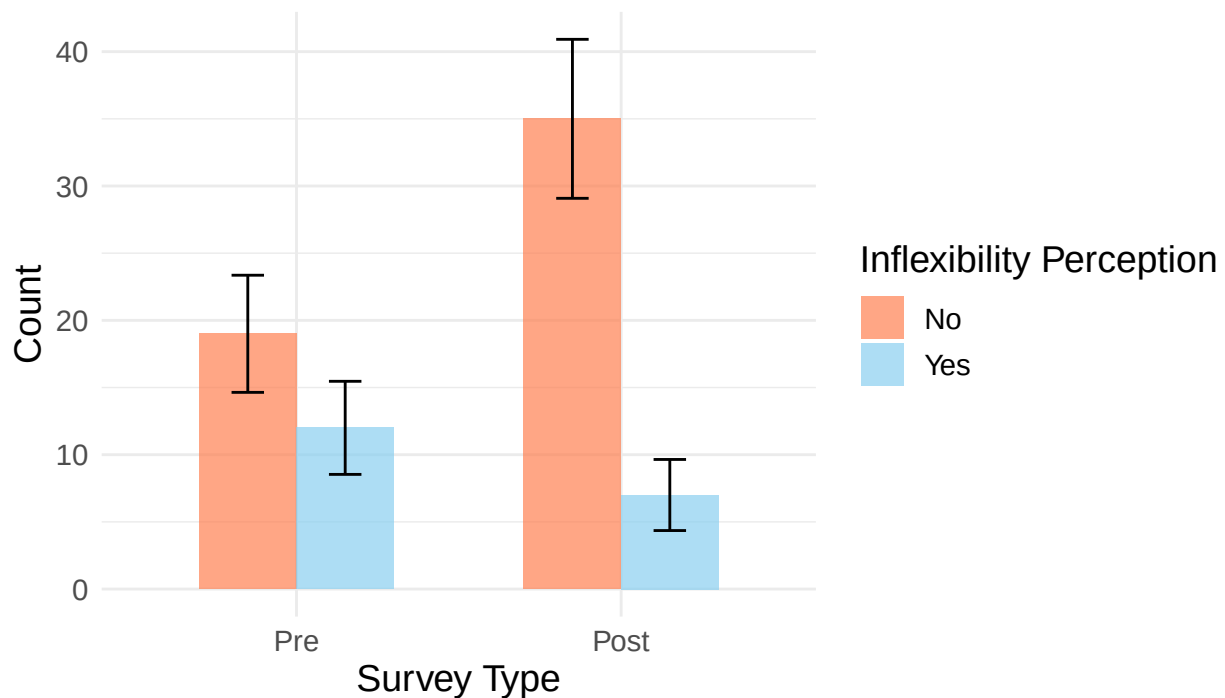
```

y = "Count",
fill = "Inflexibility Perception") +
theme_minimal() +
theme(text = element_text(size = 14))

```

Inflexibility Perception (Pre vs Post)

'Yes' means agreeing that older individuals may be inflexible; 'No' means disagreement.



```

# Read the dataset
understanding_data <- read.csv("/home/zzeng1/Innovation80/Innovation80/Understanding_Young - Sheet1.csv")

# Convert empty strings to NA and remove them
understanding_data[understanding_data == ""] <- NA
understanding_data <- understanding_data %>% drop_na(Understanding)

# Filter only Young participants
understanding_data <- understanding_data %>% filter(Participant.Type == "Young")

# Convert all "yes" to "Yes" before summarizing data
understanding_data <- understanding_data %>%
  mutate(Understanding = ifelse(Understanding == "yes", "Yes", Understanding))

# Verify the change
print(unique(understanding_data$Understanding)) # Ensure "yes" is gone

## [1] "No" "Yes"

# Create a contingency table for Understanding (Pre vs Post)
contingency_table <- table(understanding_data$Survey.Type, understanding_data$Understanding)

```

```

print(contingency_table)

##
##           No Yes
##   Post 38   4
##   Pre  17  14

# Run chi-square test
chi_test <- chisq.test(contingency_table)

# Print results
print(chi_test)

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 10.35, df = 1, p-value = 0.001295

# Relevel Survey.Type so "Pre" appears before "Post"
understanding_data$Survey.Type <- factor(understanding_data$Survey.Type, levels = c("Pre", "Post"))

# Summarize counts for each group
summary_data <- understanding_data %>%
  group_by(Survey.Type, Understanding) %>%
  summarise(Count = n(), .groups = "drop")

# Calculate standard error (SE = sqrt(n), assuming Poisson distribution)
summary_data <- summary_data %>%
  group_by(Survey.Type) %>%
  mutate(Standard_Error = sqrt(Count))

# Verify that "yes" is gone before plotting
print(unique(summary_data$Understanding))

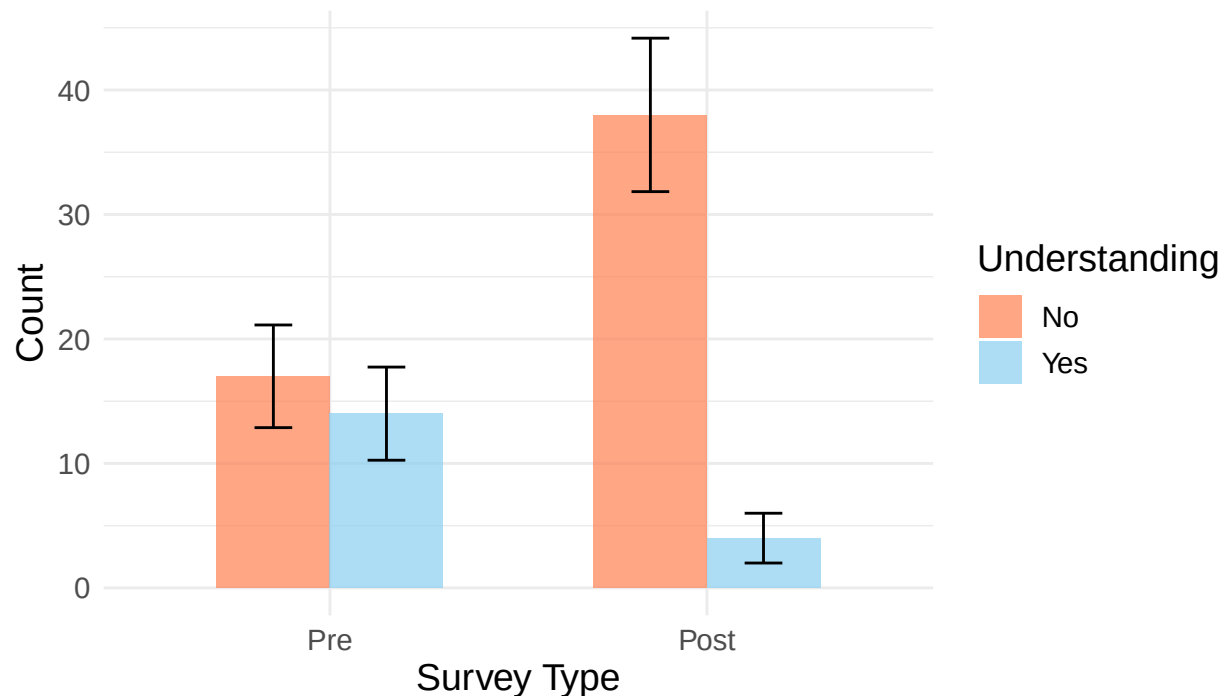
## [1] "No" "Yes"

# Plot bar chart with error bars and explanation note
ggplot(summary_data, aes(x = Survey.Type, y = Count, fill = Understanding)) +
  geom_bar(stat = "identity", position = position_dodge(), width = 0.6, alpha = 0.7) +
  geom_errorbar(aes(ymin = Count - Standard_Error, ymax = Count + Standard_Error),
    width = 0.2, position = position_dodge(0.6)) +
  scale_fill_manual(values = c("Yes" = "#89CFF0", "No" = "#FF7F50")) +
  labs(title = "Understanding (Pre vs Post) Among Young Participants",
    subtitle = "'Yes' means agreeing that older individuals
    may have difficulty understanding them; 'No' means disagreement.",
    x = "Survey Type",
    y = "Count",
    fill = "Understanding") +
  theme_minimal() +
  theme(text = element_text(size = 14))

```

Understanding (Pre vs Post) Among Young Participants

'Yes' means agreeing that older individuals may have difficulty understanding them; 'No' means disagreement



```
# Read the dataset
disinterest_data <- read.csv("/home/zzeng1/Innovation80/Innovation80/Widom_Disint - Sheet1.csv", stringsAsFactors = FALSE)

# Convert empty strings to NA and remove them
disinterest_data[disinterest_data == ""] <- NA
disinterest_data <- disinterest_data %>% drop_na(Disinterest.in.Interaction)

# Create a contingency table comparing Young vs Senior
contingency_table <- table(disinterest_data$Participant.Type, disinterest_data$Disinterest.in.Interaction)

# Run chi-square test
chi_test <- chisq.test(contingency_table)

## Warning in chisq.test(contingency_table): Chi-squared approximation may be
## incorrect

# Print results
print(chi_test)

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 0.62338, df = 1, p-value = 0.4298

# Convert empty strings to NA and remove them
disinterest_data[disinterest_data == ""] <- NA
disinterest_data <- disinterest_data %>% drop_na(Disinterest.in.Interaction)
```

```

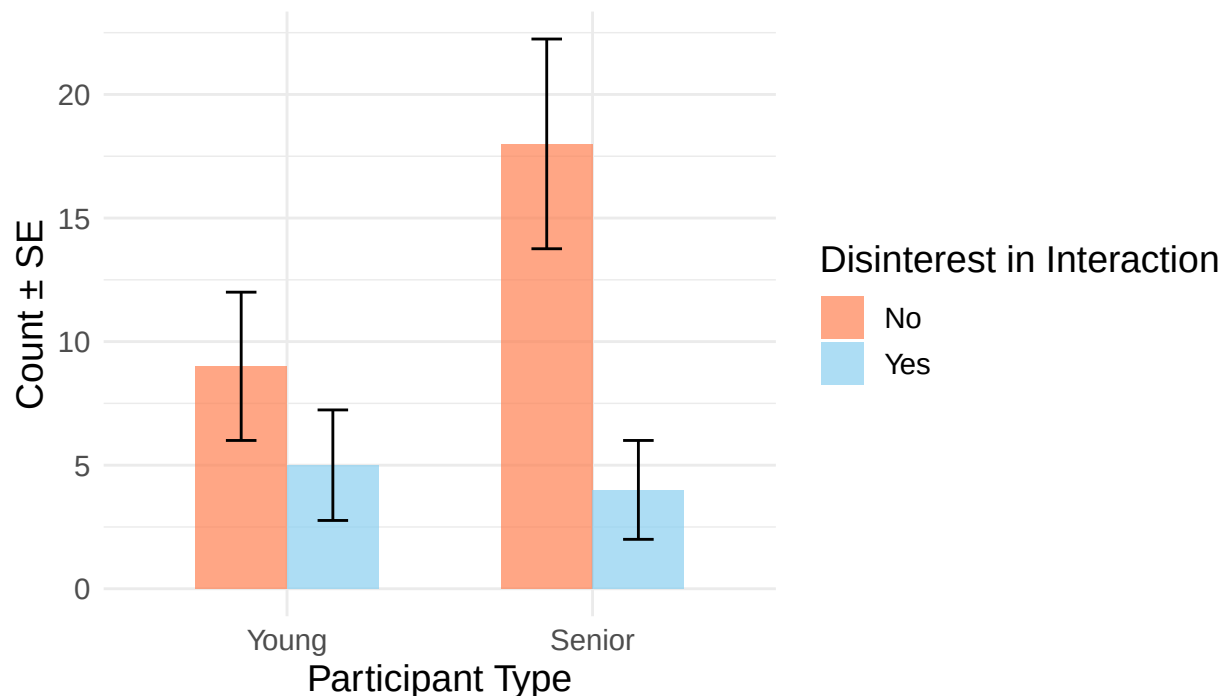
# Relevel Participant.Type so "Senior" appears before "Young"
disinterest_data$Participant.Type <- factor(disinterest_data$Participant.Type, levels = c("Young", "Senior"))

# Create a summary table with counts and standard error (SE)
summary_data <- disinterest_data %>%
  group_by(Participant.Type, Disinterest.in.Interaction) %>%
  summarise(Count = n(), .groups = "drop") %>%
  mutate(SE = sqrt(Count)) # Approximate standard error for count data

# Create the bar plot with error bars and explanation note
ggplot(summary_data, aes(x = Participant.Type, y = Count, fill = Disinterest.in.Interaction)) +
  geom_bar(stat = "identity", position = position_dodge(), width = 0.6, alpha = 0.7) +
  scale_fill_manual(values = c("Yes" = "#89CFF0", "No" = "#FF7F50")) +
  geom_errorbar(aes(ymin = Count - SE, ymax = Count + SE),
               width = 0.2, position = position_dodge(0.6)) +
  labs(title = "Disinterest in Interaction by Participant Type (Strands of Wisdom)",
       subtitle = "'Yes' means agreeing that younger/older individuals might be uninterested in interactions; 'No' means disagreement.",
       x = "Participant Type",
       y = "Count ± SE",
       fill = "Disinterest in Interaction") +
  theme_minimal() +
  theme(text = element_text(size = 14))

```

Disinterest in Interaction by Participant Type (Strands of V
 'Yes' means agreeing that younger/older
 individuals might be uninterested in interactions; 'No' means disa



```

# Read the dataset
perspective_data <- read.csv("/home/zzeng1/Innovation80/Innovation80/Widom_Perspective - Sheet1.csv", s

```

```

# Convert empty strings to NA and remove them
perspective_data[perspective_data == ""] <- NA
perspective_data <- perspective_data %>% drop_na(Perspective.Understanding)

# Create a contingency table comparing Pre vs Post
contingency_table <- table(perspective_data$Survey.Type, perspective_data$Perspective.Understanding)
print(contingency_table)

##
##           No Yes
## Post 10    0
## Pre  16    9

# Run chi-square test
chi_test <- chisq.test(contingency_table)

## Warning in chisq.test(contingency_table): Chi-squared approximation may be
## incorrect

# Print results
print(chi_test)

##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 3.1448, df = 1, p-value = 0.07617

# Convert empty strings to NA and remove them
perspective_data[perspective_data == ""] <- NA
perspective_data <- perspective_data %>% drop_na(Perspective.Understanding)

# Relevel Survey.Type so "Pre" appears before "Post"
perspective_data$Survey.Type <- factor(perspective_data$Survey.Type, levels = c("Pre", "Post"))

# Create a summary table with counts and standard error (SE)
summary_data <- perspective_data %>%
  group_by(Survey.Type, Perspective.Understanding) %>%
  summarise(Count = n(), .groups = "drop") %>%
  mutate(SE = sqrt(Count)) # Approximate standard error for count data

# Create the bar plot with error bars and explanation note
ggplot(summary_data, aes(x = Survey.Type, y = Count, fill = Perspective.Understanding)) +
  geom_bar(stat = "identity", position = position_dodge(), width = 0.6, alpha = 0.7) +
  scale_fill_manual(values = c("Yes" = "#89CFF0", "No" = "#FF7F50")) +
  geom_errorbar(aes(ymin = Count - SE, ymax = Count + SE),
               width = 0.2, position = position_dodge(0.6)) +
  labs(title = "Perspective Understanding by Survey Type (Strands of Wisdom)",
       subtitle = "'Yes' means agreeing that younger/older
       individuals might struggle to understand different perspectives; 'No' means disagreement.",
       x = "Survey Type (Pre/Post)",
       y = "Count ± SE",
       fill = "Perspective Understanding") +
  theme_minimal() +
  theme(text = element_text(size = 14))

```

Perspective Understanding by Survey Type (Strands of W

'Yes' means agreeing that younger/older individuals might struggle to understand different perspectives; 'N

